

CS228 : Logic for Computer Science

Module 3: Predicate Logic

Lecture 13: Semantics of Predicate Logic

Instructor: S. Akshay

IIT Bombay, India

Syllabus for Mid-Semester Exam

- ▶ Module 1: Propositional Logic
- ▶ Module 2: The Problem of Satisfiability
- ▶ All lectures till and including Lecture 11

Mid-Semester Exam

- ▶ Venue: LA 201, LA 002; PwD students at KR 125
- ▶ Seating arrangement will be shared on Piazza
- ▶ 1 A4 size paper allowed as cheatsheet, but **MUST** be handwritten with blue ink – no photocopy, etc.

Topics Covered till date

Module 1: Propositional Logic

1. Syntax and Semantics
2. Formal proof systems
3. Normal Forms

Module 2: The Problem of Satisfiability

1. Propositional Resolution - soundness, termination and completeness
2. The DPLL algorithm
3. Clause learning, the CDCL algorithm, and modern SAT solvers
4. A tool demo of SAT/SMT solvers in practice

Module 3: Predicate Logic

1. Syntax of predicate logic (Lecture 12)
2. Today: **semantics of predicate logic**

Quick Recap: Syntax of Predicate Logic

- ▶ **Terms:** $t ::= x \mid c \mid f(t, \dots, t)$
where x is a variable, c a constant, f a function symbol
- ▶ **Formulas:** $\varphi ::= P(t_1, \dots, t_n) \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \varphi \rightarrow \varphi \mid \forall x \varphi \mid \exists x \varphi$
where P is a predicate symbol
- ▶ **Free/bound variables:** $free(\varphi)$, $bnd(\varphi)$ defined recursively – not complements of each other!
- ▶ **Substitution:** $\varphi[t/x]$ replaces free occurrences of x by t – but only if t is **free for x in φ** (no variable of t gets captured by a quantifier in φ)!

Today: what do these formulas actually *mean*?

Topic 13.1

Semantics of Predicate Logic

Semantics of FOL

Example 13.1 $\psi_1 := \forall x \forall y (P(x, y) \rightarrow \exists z (\neg(z = x) \wedge \neg(z = y) \wedge P(x, z) \wedge P(z, y)))$

1. What does this mean? English meaning: For every x, y if $P(x, y)$ holds, then we can find z different from x, y such that $P(x, z), P(z, y)$ hold.
2. Is the statement true?
 - ▶ Case 1: Suppose,
 - ▶ variables take values from **real numbers**.
 - ▶ $P(x, y)$ represents $x < y$
 - ▶ **Then, ψ_1 is true, i.e., real numbers are dense!**
 - ▶ Case 2: Instead suppose,
 - ▶ variables take values from **real numbers**, but $P(x, y)$ represents $x \leq y$
 - ▶ **Then, ψ_1 is false!**
 - ▶ Case 3: Finally, suppose,
 - ▶ variables take values from **natural numbers**, with $P(x, y)$ representing $x < y$
 - ▶ **Then, ψ_1 is false, since natural numbers are not dense.**

Truth value of any formula φ depends on (i) **underlying set** from which variables take values and (ii) **interpretation of constants, functions and predicates, i.e., Vocabulary!**

Semantics of FOL

- ▶ Vocabulary $\mathcal{V}$. E.g., $\mathcal{V} = \{a, f, =, P\}$
- ▶ $\mathcal{V}$ -formula. E.g., $\psi_2 := \exists x(P(x, f(y, a)) \wedge \neg(x = a))$, $var(\psi_2) = \{x, y\}$, $free(\psi_2) = \{y\}$.

Truth of φ depends on

1. Universe U from which variables take values, e.g., $\mathbb{N}$
2. Interpretation of $\mathcal{V}$ on U
 - ▶ Each constant to an element of U , e.g., $a \mapsto 0$
 - ▶ Each n -ary function symbol to a function from U^n to U , e.g., $f(u, v) = u + v$.
 - ▶ Each m -ary predicate symbol to a subset of U^m , e.g.,
 - ▶ Interpret for $=$: $\{(c, c) \mid c \in \mathbb{N}\}$ - **fixed interpretation!**
 - ▶ Interpret for P : $\{(c, d) \mid c, d \in U, c < d\}$

This defines a $\mathcal{V}$ -structure (or Model) $M = (U^M, (a^M, f^M, P^M))$, e.g., $(\mathbb{N}, (0, +, <))$

3. We also need an environment $\alpha : var(\varphi) \rightarrow U$, e.g., $\alpha(y) = 2, \alpha(x) = 1$.
 - ▶ We will also use $\alpha[x \mapsto u]$ to denote the env which maps x to u (other vars are mapped according to α).

Semantics of FOL: An Example First

- ▶ **Example 13.2** Consider again $\psi_2 := \exists x(P(x, f(y, a)) \wedge \neg(x = a))$ over Vocabulary $\mathcal{V} = \{a, f, =, P\}$
- ▶ Also, let $M = (\mathbb{N}, (0, +, <))$ be a $\mathcal{V}$ -structure, and $\alpha(x) = 0, \alpha(y) = 2$ be the env.
- ▶ Does M, α satisfy ψ_2 ?
 - ▶ First compute $f(y, a) = y + a = 2 + 0 = 2$ (using $\alpha(y) = 2, a^M = 0$).
 - ▶ Is $P(x, f(y, a))$ true, i.e. is $x < 2$?
 - ▶ Is $\neg(x = a)$ true, i.e. is $x \neq 0$?
 - ▶ Is their conjunction true, i.e. is $x < 2$ and $x \neq 0$?
 - ▶ Finally, is there a value of x from the universe $\mathbb{N}$ that makes their conjunction true?
 - ▶ With $x = 0$ it is not satisfied.
 - ▶ **But, $x = 1$ works!** And that is all we need.
 - ▶ In other words, the env α , where we set $x = 1$ (ignoring the earlier value) satisfies the formula, since $(1 < 2 \wedge 1 \neq 0)$ is true!
 - ▶ Thus, we can conclude $M, \alpha \models \psi_2$!

Now, let's make this definition precise and generic!

Semantics of FOL: Formal Definition

Given $\mathcal{V}$ -structure/model M and env α , we define $M, \alpha \models \varphi$ (satisfaction relation)

By structural induction! First we extend α to $\bar{\alpha} : \text{Terms}(\varphi) \rightarrow U^M$, by

- ▶ If t is a variable x , then $\bar{\alpha}(t) = \alpha(x)$
- ▶ If t is $f(t_1, \dots, t_m)$, then $\bar{\alpha}(t) = f^M(\bar{\alpha}(t_1), \dots, \bar{\alpha}(t_m))$
In [Example 13.2](#), we have $\bar{\alpha}(f(y, a)) = f^M(\alpha(y), a^M) = 2 + 0 = 2$

- ▶ $M, \alpha \models P(t_1, \dots, t_m)$ iff $(\bar{\alpha}(t_1), \dots, \bar{\alpha}(t_m)) \in P^M$ (atomic formulas)
- ▶ $M, \alpha \models \neg\varphi$ iff $M, \alpha \not\models \varphi$.
- ▶ $M, \alpha \models \varphi_1 \wedge \varphi_2$ iff $M, \alpha \models \varphi_1$ and $M, \alpha \models \varphi_2$.
- ▶ $M, \alpha \models \exists x\varphi$ iff there exists some $u \in U^M$ s.t., $M, \alpha[x \mapsto u] \models \varphi$.
- ▶ $M, \alpha \models \forall x\varphi$ iff for all $u \in U^M$, $M, \alpha[x \mapsto u] \models \varphi$.

We omit $\vee, \rightarrow, \Leftrightarrow$ here since they reduce to $\wedge, \neg$ in the usual way (e.g. $\varphi_1 \rightarrow \varphi_2 \equiv \neg\varphi_1 \vee \varphi_2$) – same treatment as $\wedge$ above. **Exercise 13.1** In the e.g., what if we chose (i) $\alpha(y) = 0$? (ii) $\alpha(x) = 5$? (iii) $\alpha(x) = 1$?

Satisfaction only depends on the “free” environment

Exercise 13.2

Show that if α, α' are identical on the set of free variables, then for any formula φ and any $\mathcal{V}$ -structure M , $M, \alpha \models \varphi$ iff $M, \alpha' \models \varphi$.

hint: use structural induction!

- Thus, if φ has no free variables, i.e., it is a sentence, then α does not matter!

For a sentence φ , we just write $M \models \varphi$.

Recall: Truth Depends on the Structure

In Example 13.1 $\psi_1 := \forall x \forall y (P(x, y) \rightarrow \exists z (\neg(z = x) \wedge \neg(z = y) \wedge P(x, z) \wedge P(z, y)))$

- ▶ For structure $M_1 = (\mathbb{R}, <)$: $M_1 \models \psi_1$ (density!)
- ▶ For structure $M_2 = (\mathbb{R}, \leq)$: $M_2 \not\models \psi_1$
- ▶ For structure $M_3 = (\mathbb{N}, <)$: $M_3 \not\models \psi_1$

What is the vocab here?
Why is there no α ?

In Example 13.2 $\psi_2 := \exists x (P(x, f(y, a)) \wedge \neg(x = a))$ over $\mathcal{V} = \{a, f, =, P\}$

- ▶ For $M_4 = (\mathbb{N}, (0, +, <))$ with $\alpha(x) = 0, \alpha(y) = 2$: $M_4, \alpha \models \psi_2$ (witness $x = 1$)
- ▶ For $M_4 = (\mathbb{N}, (0, +, <))$ with $\alpha'(x) = 0, \alpha'(y) = 0$: $M_4, \alpha' \not\models \psi_2$ (need $x < 0$, no natural number works!)

Exercise 13.3 Find a *different* structure M' (and/or environment α'') for formula ψ_2 , and determine whether $M', \alpha'' \models \psi_2$.

What happens when $M' = (\mathbb{N}, (1, \times, <))$? Is there an env which makes the formula true/false?

Revisiting substitution

Exercise 13.4 Let $\psi_3(x) := \exists y (y > x)$ over $\mathcal{V} = \{>\}$, with $M = (\mathbb{N}, >)$.

1. What is $\text{free}(\psi_3)$? For any α , does $M, \alpha \models \psi_3$ hold?
2. Is $t = y$ free for x in ψ_3 ? If we substitute anyway, what is $\psi_3[y/x]$? What does this tell us about the substitution definition?

$\text{free}(\psi_3) = \{x\}$. For *any* α , is there $u \in \mathbb{N}$ with $u > \alpha(x)$? **Yes, always!** Numbers are unbounded above, so $M, \alpha \models \psi_3$ for every α .

Recall **Substitution in FOL**: $t = y$ is not free for x in ψ_3 : (y occurs in t , and the free x is under the scope of $\exists y$)

- Substituting anyway, step by step:

$$\exists y (y > x) \longrightarrow \exists y (y > \textcolor{red}{y})$$

- So $\psi_3[y/x] = \exists y (y > y)$
- Is *this* true? **No – always false!** No number is bigger than itself.

The truth value flipped. This is exactly why substitution is restricted to terms that are free for x .

Semantic entailment

Once we fix the vocab $\mathcal{V}$, we just write structure instead of $\mathcal{V}$ -structure.

Let Γ be a (possibly infinite) set of formulas and let ψ be a formula in predicate logic.

- ▶ **Semantic entailment** $\Gamma \models \psi$ holds iff for all M and α , whenever $M, \alpha \models \varphi$ holds for all $\varphi \in \Gamma$, then $M, \alpha \models \psi$ holds as well.
- ▶ ψ is **satisfiable** iff there is some M and α such that $M, \alpha \models \psi$ holds.
- ▶ ψ is **valid** iff $M, \alpha \models \psi$ holds for all M and α .
- ▶ Γ is **consistent** iff there is at least one M and α such that $M, \alpha \models \varphi$ for all $\varphi \in \Gamma$.

Note: M ranges over *all* universes and *all* interpretations of $\mathcal{V}$ on them – not a fixed universe!

Exercise 13.5 Give 2 examples of valid and 2 of satisfiable (but not valid) formulas with the vocab given in Example 13.2.

Exercise 13.5 Solution

Vocab: constant a , unary function f , binary predicate P (plus always-present $=$).

Valid (true for every M and α , regardless of interpretation)

1. $a = a$ – equality has a *fixed* interpretation (always literal identity), so this holds no matter how a is interpreted.
2. $\forall x (P(x, f(a)) \rightarrow P(x, f(a)))$ – an instance of the propositional tautology $\varphi \rightarrow \varphi$, lifted into predicate logic; true regardless of how P, f, a are interpreted.

Satisfiable but *not* valid

1. $P(a, a)$ – true if we pick M with $P^M = U \times U$ (P always holds); **false** in our familiar $M_4 = (\mathbb{N}, (0, +, <))$ since $0 < 0$ is false. Satisfiable, not valid.
2. $\exists x P(x, a)$ – true if $U = \mathbb{Z}$, P interpreted as “ $<$ ”, $a = 0$ (witness $x = -1$); **false** in $M_4 = (\mathbb{N}, (0, +, <))$ since no natural number is < 0 . Satisfiable, not valid.

Semantic equivalences

Definition

$\varphi \equiv \psi$ iff $\{\varphi\} \models \psi$ and $\{\psi\} \models \varphi$.

Exercise 13.6 Prove the following: Quantifier equivalences

- ▶ $\forall x\varphi \equiv \neg\exists x\neg\varphi$
- ▶ $\forall x\forall y\varphi \equiv \forall y\forall x\varphi$
- ▶ $\forall x(\varphi_1 \wedge \varphi_2) \equiv (\forall x\varphi_1) \wedge (\forall x\varphi_2)$
- ▶ $\exists x(\varphi_1 \vee \varphi_2) \equiv (\exists x\varphi_1) \vee (\exists x\varphi_2)$